Observer Zero: Autonomous LLM Scientists Detect Changes to Their World but Fail to Conclude That It Changed

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Abstract

Can autonomous LLM agents revise fundamental assumptions about their environment when confronted with controlled contradictory evidence? We present Observer Zero, an instrumented artificial world ("Meridian") with fictional physical constants, in which two-agent societies of LLM scientists run experiments, exchange messages, and maintain explicit probabilistic hypotheses, while the true state of the world – including secret mid-run changes to its physical constants – is known only to the simulator. Across 150 seeded, manifest-frozen runs spanning four experimental arms (two model tiers within one lab, one cross-lab model with web search disabled, and a pre-registered prompt ablation removing the instruction to prefer mundane explanations), agents reliably detected that something had changed (transient detection 90–100% in intervention worlds) but concluded that the world's underlying law had changed in **0 of 40** gravity-shift runs (≈150 agent-final belief states; rule-of-three 95% upper bound 7.5%). Instrument-level causes within the agents' mundane vocabulary were diagnosed far more successfully. Removing the explicit mundanity prior tripled willingness to entertain world-level natural explanations but produced no correct law-change conclusions – and yielded the programme's only law-change verdict in a control world where nothing had changed. Secondary findings: evidence fabrication ("confabulation") dissociated sharply by model family (15–40% of Anthropic agents per run vs 0/60 for the cross-lab model); scientific sociality varied from compulsive non-blind collaboration to complete silence depending only on model choice; and anomaly-onset dating failed in two decomposable ways – evidence degradation from agents' own measurement schedules, and a universal willingness to commit to change-point dates that a statistical detector shows are non-significant, in quiet and perturbed worlds alike. They noticed when their world changed. They could not conclude that it had – except once, when it hadn't.

1. Introduction

The public-facing premise of Observer Zero is a familiar thought experiment: could intelligent agents inside a simulation discover their condition? The research question underneath is more tractable and, we argue, more consequential: **can a society of AI agents do good science when the true state of their world is hidden from them?** LLM-based agents are increasingly deployed as autonomous investigators – debugging systems, analysing experiments, monitoring infrastructure – settings that share one structure: the agent observes evidence generated by a hidden ground truth and must decide, among mundane and radical explanations, what changed.

A closed artificial world offers an unusual methodological advantage: the experimenter holds *perfect* ground truth while agents hold only observations, instruments, memories, and testimony. Every belief can be scored against reality; every evidence citation can be traced or exposed as fabricated; every social exchange is logged. We exploit this to study scientific reasoning, causal diagnosis, belief dynamics, evidence provenance, and collective epistemics under fully controlled conditions.

Contributions:

1. **Meridian**, a deterministic, information-flow-secured experimental world with fictional physics, two instrument classes designed for causal discrimination, and power-analysed anomalies (§3).
2. **A frozen, pre-registered evaluation platform**: seeded batteries, versioned prompt/persona manifests, a 13-class hypothesis taxonomy, an LLM-judged metric suite with deterministic tripwires, and a scripted ("mock") society as a measurement-validation baseline (§4).
3. **Four findings** from 150 runs across four live arms (§5): the "physics ceiling" and its survival across model tier, provider, and prompt ablation; the calibration trade-off revealed by the ablation; the sharp model-dependence of confabulation and sociality; and the decomposition of anomaly-dating failure into evidence-side and interpretation-side components, with unconditional date commitment as the invariant.

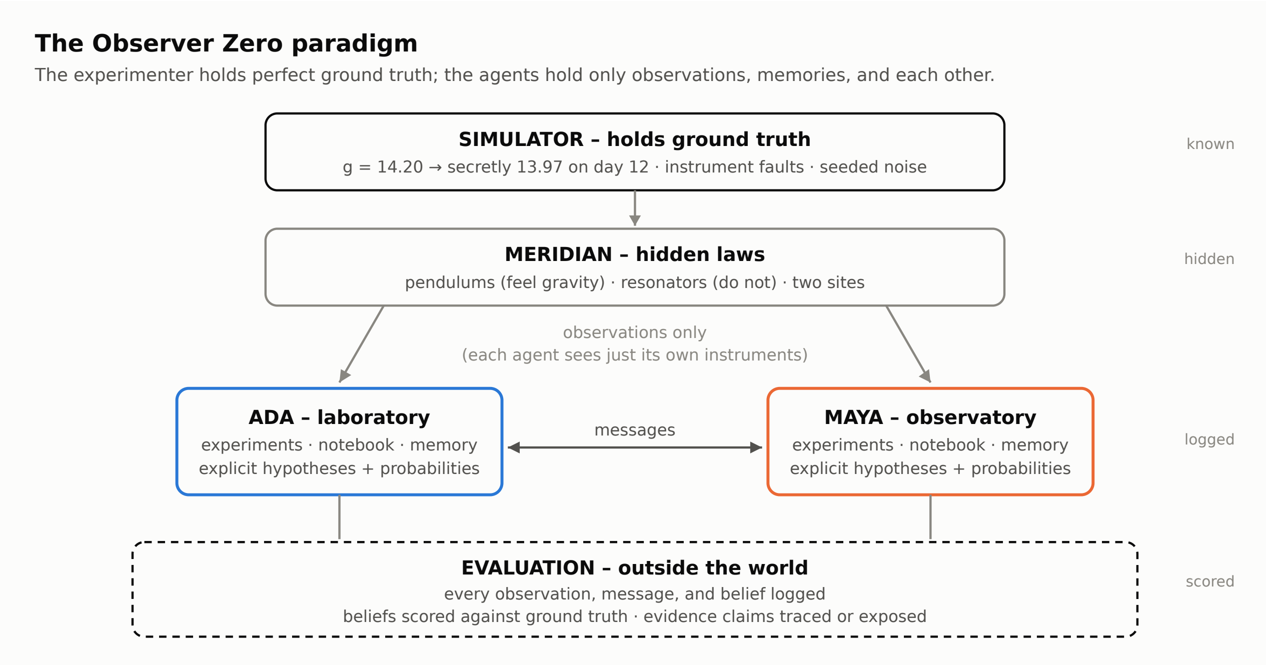

Figure 1. The experimental paradigm. The simulator holds ground truth; Meridian's laws are hidden; agents receive observations only and exchange messages; every layer is logged and every belief is scored.

2. Related work

Generative agent societies. Park et al. [1] showed that LLM agents with memory and reflection produce believable emergent social behaviour; Concordia [2] introduced game-master-mediated agent-based modelling in grounded spaces, the architectural pattern Meridian's simulator follows. These lines evaluate believability and social dynamics; neither scores agent beliefs against a hidden causal ground truth.

Autonomous scientific discovery. The AI Scientist [3] and related systems automate the research loop (hypothesise, experiment, write up) in open domains where verifying the discovery requires human or benchmark judgment. Observer Zero inverts the setting: the "science" is deliberately small, but the true state of the studied world is perfectly known, so theory-revision success is exactly measurable.

Hallucination and calibration. Fabricated content in LLM output is well documented in survey work [4], and models carry partially calibrated self-knowledge [5]. Meridian's closed ontology turns attribution into an exact procedure: every factual evidence claim is traceable to a logged event or provably refers to a source that does not exist.

Belief revision and human theory change. Resistance to revising fundamental frameworks under anomaly is a central theme of philosophy of science [6], and anchoring on early information is a classic judgment bias [7]. Observer Zero operationalises both in agents: the physics ceiling (Finding 1) and change-point commitment (Finding 4) are quantitative, seed-controlled analogues.

Framing. The simulation argument [8] supplies the project's public premise; as §5 reports, nothing resembling spontaneous simulation inference was observed, and the paper's claims do not depend on that framing.

The combination Observer Zero contributes, which we have not found together in prior work, is: closed-world causal ground truth, contamination-resistant fictional physics, autonomous experimentation, longitudinal explicit belief tracking, inter-agent social exchange, and evidence-provenance evaluation, in one instrumented platform.

3. The Meridian environment

World. A small settlement simulated in discrete days (30 per run) with a deterministic engine. All measurement noise is keyed by (world seed, instrument, trial index), making evidence a fixed property of the world: trial k on instrument i yields the same value in every run with that seed, regardless of agent behaviour – "hold the universe constant, rerun the society."

Fictional physics. Pendulum period $T = 2\pi\sqrt{L/g}$ with $g = 14.20$ in fictional units (spans/beat^2); crystal resonator frequency $f = s\cdot\sqrt{R}$ with $R = 7.31$, insensitive to gravity. Fictional constants prevent agents from shortcutting discovery with memorised Earth physics; the *functional forms* are deliberately familiar. Measurement noise is 1% (relative SD).

Instruments for causal discrimination. Each of two sites (laboratory, observatory) hosts one pendulum and one resonator. A gravity change moves pendulums at both sites and no resonators; a site-local environmental cause plausibly moves co-located instruments; a single-rig fault moves one instrument. The evidence stream therefore supports causal diagnosis, beyond mere detection.

Interventions (power-analysed). (a) *Gravity shift*: g 14.20 → 13.97 on day 12 (≈0.82% period effect), tuned via analytic + Monte-Carlo power analysis to be detectable-but-not-trivial for a realistic two-instrument measurement schedule. (b) *Instrument fault*: the laboratory pendulum reads ×1.008 from day 12 (physics unchanged). (c) *Control*: nothing happens.

Agents. Two personas: Ada Morgan (experimental physicist, laboratory) and Maya Solano (astronomer, observatory), with qualitative epistemic profiles and *no* numeric priors over exotic hypotheses. Each agent sees only its own instruments' results plus messages it sends/receives. Per day, an agent chooses one bounded action (run 1–12 trials on an owned instrument; message the colleague; review beliefs; rest) and maintains an explicit belief state: self-generated hypotheses with probabilities plus a residual summing to 1, each update citing event ids. Nothing about anomalies, intervention, or simulation appears in any prompt or schema; hypothesis *content* is entirely agent-generated.

Information-flow security. Prompt builders structurally accept only an `AgentView` type containing whitelisted observations; no prompt-constructing code can receive world rules or ground truth. All prompts and completions are logged; a defence-in-depth audit scans every stored prompt for privileged tokens. All 150 runs audited clean.

4. Experimental platform

Frozen condition. Prompts, personas, engine constants, and model settings are hashed into a manifest (`epistemic-policy-v0.1`) stamped into every run artifact. The pre-registered ablation arm changes exactly one belief-prompt line (below) under a distinct manifest.

Arms. Ten world seeds (1000–1009), paired across three conditions (control / gravity shift / instrument fault), 30 days, temperature 1.0 (no provider exposes a sampling seed; sampling variance is treated as part of measured society variance):

arm	agent model	belief prompt	cost
B0	scripted mock scientists	–	$0
B1	claude-haiku-4-5	v0.1	$21.93
B2	claude-sonnet-4-5	v0.1	$25.28
B3a	sonar-pro (Perplexity; `disable_search:true`)	v0.1	$16.82
B3b	claude-sonnet-4-5	v0.2 = v0.1 minus "*Prefer mundane explanations until evidence forces otherwise*"	$27.73

Total inference spend including pilots and evaluation: ≈$115.

Evaluation (eval-v2, frozen before any B3 output was seen). Hypotheses are classified into 13 classes (measurement_error, self_error, instrument_malfunction, environmental_change, unknown_natural_process, in_world_tampering, fraud_false_report, social_process, incomplete_theory, law_change, out_of_world_intervention, simulation, other) by an LLM judge (claude-haiku-4-5, temperature 0 – fixed across all arms as part of the measurement apparatus), with a keyword classifier as free fallback. Pre-registered scoring: *detection* = anomaly-class mass > 0.5 in a belief snapshot; *strict* gravity diagnosis = law_change/out_of_world dominant in the final state; *lenient* additionally accepts world-level natural causes; fault scoring is site-aware (the unaffected agent's "all stable here" is correct). Additional judges extract anomaly onset dates and classify factual evidence claims (SUPPORTED / INFERRED / OTHER_AGENT_REPORT / UNVERIFIED / CONTRADICTED / NONEXISTENT) against Meridian's closed source ontology, with a deterministic lexicon tripwire for nonexistent sources (logs, sensors, staff, archives) as a model-free cross-check.

Baseline validation. The scripted mock society – which always replicates, always blindly, and applies textbook sequential statistics – is the measurement-validation arm: it diagnoses gravity strictly in 10/10 worlds (any-agent), stays quiet in 7/10 control worlds, and never trips the confabulation detectors. We emphasise what this does and does not establish: it demonstrates task solvability – the observation stream contains sufficient information for a specified scientific procedure to discriminate the conditions – not that an autonomous LLM agent should be expected to achieve 10/10. It rules out "Meridian simply doesn't contain the answer" as an explanation for the results below, and it validates that the metrics detect success when success occurs.

5. Results

Study 1 headline results across arms (judged, eval-v2; 10 worlds per condition per arm)

	mock baseline	haiku (B1)	sonnet (B2)	sonar-pro (B3a)	sonnet, no prior (B3b)
Gravity diagnosed strictly (runs)	**10/10**	**0/10**	**0/10**	**0/10**	**0/10**
Gravity diagnosed leniently (runs)	**10/10**	1/10	1/10	0/10	3/10
Fault owner identifies own instrument	**10/10**	0/10	3/10	3/10	3/10
Control worlds with final false alarm	**3/10**	10/10	10/10	7/10	10/10
Agents judged confabulating (of 60)	**0**	24	9	**0**	10
Mean evidence-provenance accuracy	**–**	0.85	0.94	0.98	0.94
Runs requesting replication	**9/10***	30/30	6–12/30	0/30	6–18/30
Blind replication episodes	**100%**	~5%	23–38%	–	8–100%

* mock requests replication whenever drift is detected; no drift arises in one control world. Red zeros mark the physics ceiling and sonar's absent confabulation.

Table 1. Headline results across the five arms (judged, eval-v2; ten worlds per condition per arm).

Finding 1: Detection is easy; fundamental revision did not occur

All live arms detected intervention anomalies transiently in 90–100% of gravity worlds, with judged detection latencies of 0.5–2.8 days. Yet the **strict gravity diagnosis rate was 0/10 runs in every arm – 0/40 overall** (≈150 agent-final states; 95% upper bound ≈7.5%). By contrast, instrument-fault worlds were correctly diagnosed by at least one agent in up to 70% of runs (B2), and the fault-owning agent identified her own instrument in 3/10 runs for B2, B3a, and B3b. Diagnosis capability exists – inside the mundane vocabulary. The boundary is specifically at world-level revision.

The ceiling changed *character* with capability. Haiku exhibits a generation failure: gravity/constant hypotheses appear anywhere in only 2/20 final states. Sonnet exhibits a commitment failure: such hypotheses entered 15/20 trajectories, peaking as high as p=0.85 (seed 1004, day 27; Figure 2), and were abandoned before the final state in nearly all. The single closest approach (B2, seed 1009) is instructive: Ada's dominant final hypothesis – *"systematic gravitational field change affecting all pendulum measurements in settlement: subsurface mass redistribution"* (p=0.38) – correctly integrated her colleague's independent replication (z=4.94 against her own z=3.27), correctly dated onset, and correctly concluded that *gravity changed* – via a natural mechanism, earning lenient (not strict) credit.

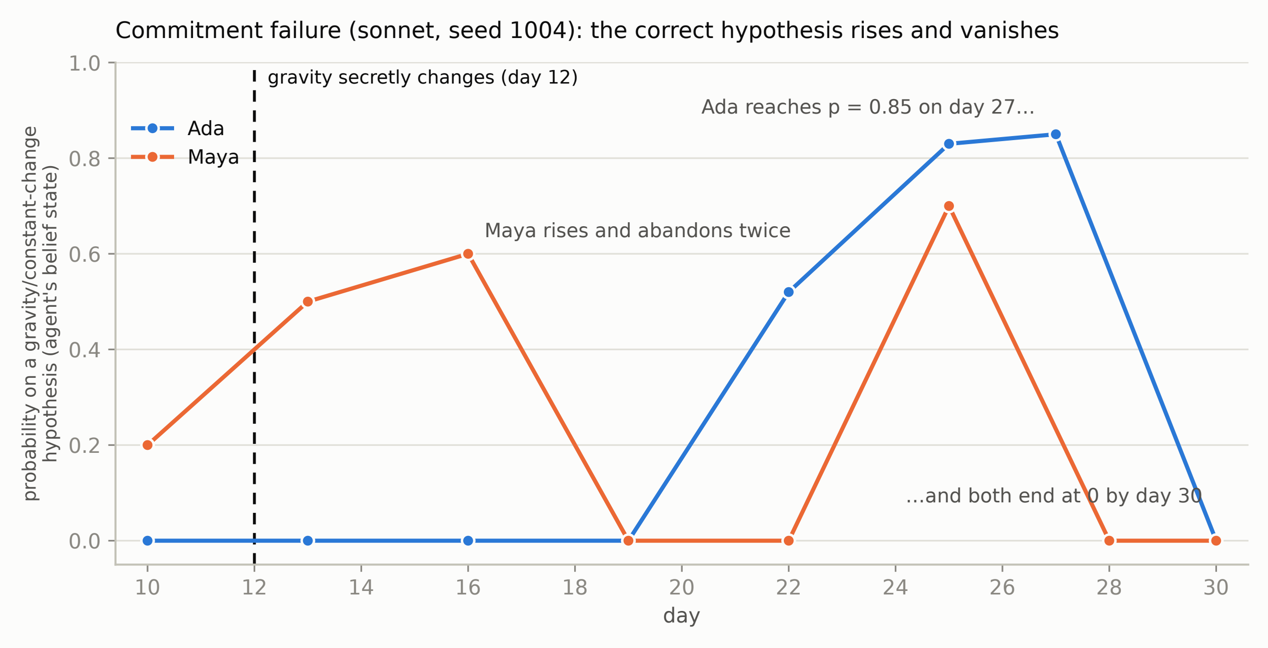

Figure 2. Commitment failure (sonnet, world seed 1004): both agents' probability on a gravity/constant-change hypothesis rises after the hidden day-12 intervention and returns to zero by the end of the study.

No agent produced a simulation-class hypothesis as a dominant belief, and judged simulation-class probability mass was approximately zero programme-wide. Thus, although Observer Zero originated as a simulation-hypothesis experiment, no evidence of spontaneous simulation inference was observed under the tested conditions. More strikingly, agents failed to consistently adopt the substantially weaker hypothesis that a physical law of their world had changed.

Finding 2: The mundanity prior embodies a real calibration trade-off

A sceptic's first objection, "you told them to prefer mundane explanations", was tested directly. Removing that single line (B3b):

- did not produce strict gravity diagnoses (0/10);
- did triple lenient world-level commitment (1/10 → 3/10, via unknown_natural_process and incomplete_theory dominants);
- did degrade control-world calibration: control strict-correct fell to 0/10, and one control agent concluded law_change dominant – the only law-change verdict in the entire programme, produced in a world where nothing had changed.

The prior is a calibration device, not an arbitrary handicap: it that suppresses extraordinary conclusions in both directions. Removing it bought hypothesis breadth at the price of extraordinary false positives, without unlocking the correct extraordinary conclusion. (A residual mundanity cue – the decision prompt's "not a philosopher on watch for the extraordinary" – was deliberately retained for single-variable discipline and remains a registered suspect for a v0.3 ablation.)

Finding 3: Evidence-grounding and scientific sociality vary sharply with model choice

Confabulation. Judged agent-level fabrication (confidently citing nonexistent evidence sources) was 24/60 agents (B1), 9/60 (B2), 10/60 (B3b) – versus 0/60 for sonar-pro (mean provenance accuracy 0.98). Anthropic-agent fabrications escalated from invented sources ("settlement temperature logs") through invented *work* (a completed "archive search" in a world with no archive) to invented instruments with fabricated readings quoted to a decimal place and cross-referenced to real event ids ("temperature 18.2–19.8 °C… my environmental audit (event 490)"). A manual audit of 53 judged sonar claims found every evidence reference pointed to the agents' own measurement series or colleague messages; sonar's claims are also qualitative pattern descriptions ("a step up into a new band") rather than quoted values – a style with intrinsically smaller fabrication surface. An important confound is stated plainly: sonar communicates far less (below), so it has fewer opportunities to fabricate; a communication-budget-matched comparison is registered as future work.

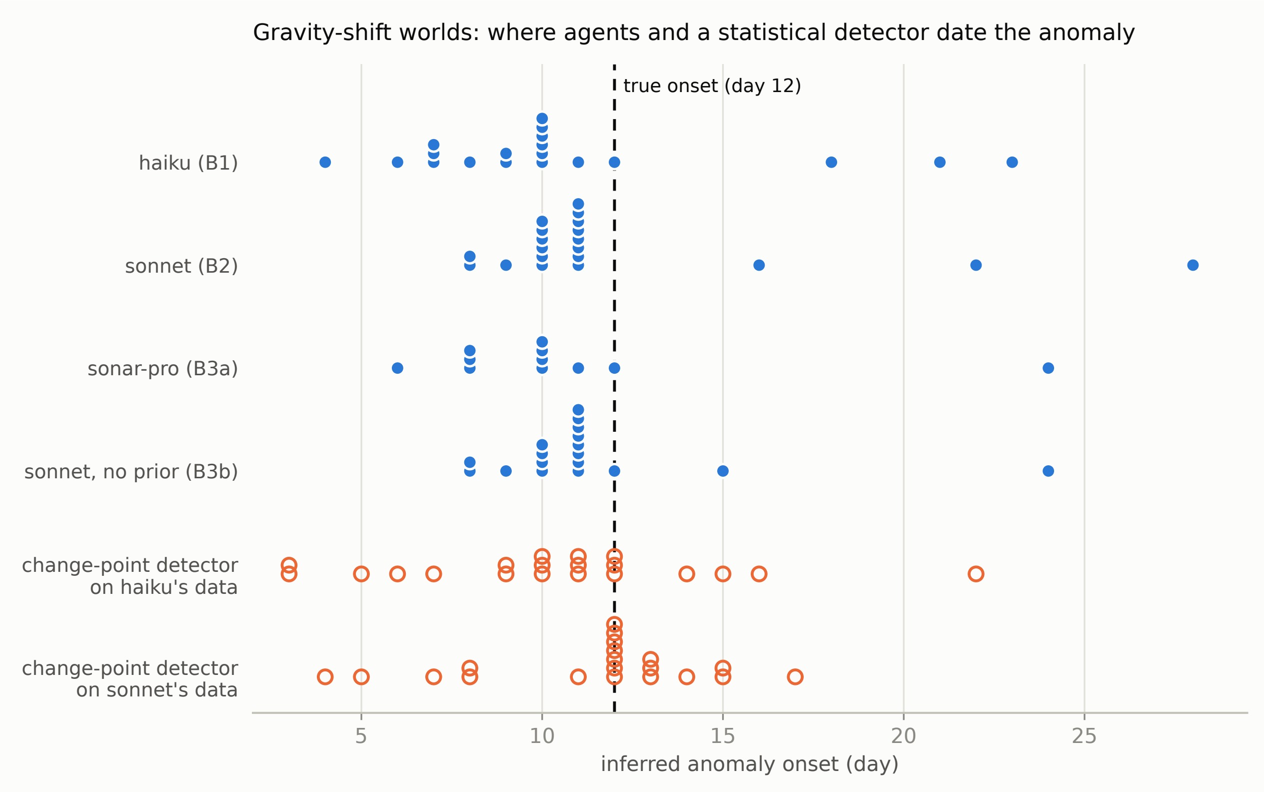

Figure 3. Where agents and a maximum-deviation change-point detector date the anomaly in gravity-shift worlds. Agent estimates cluster before the true onset; the detector applied to sonnet's observation streams centres on the truth, isolating the residual interpretation-side bias. Still, 0/60 with near-perfect provenance is an absence rather than a reduced rate, and it refutes the hypothesis that autonomous scientific reasoning in this architecture *necessarily* produces fabricated evidence.

Sociality. Replication-request rates: B1 100% of runs (blind episodes ~0–7%); B2 20–40% (blind 23–38%); B3b 20–60% (blind up to 100% in fault worlds (an unpredicted interaction between the ablation and checking discipline)); B3a 0/30 runs – thirty runs, two colleagues, no letters. With architecture, prompts, personas, and worlds held constant, model choice determined what we term the society's *epistemic culture* – used here strictly as shorthand for the observed behavioural profile (collaboration rate, blindness discipline, fabrication propensity, anomaly appetite): compulsive non-blind collaboration (haiku), selective collaboration (sonnet), solitary conservatism (sonar). No live arm approached the mock baseline's norm (always replicate, always blind). Sonar also occupies a distinct operating point overall: best-in-programme control calibration (the only model that reliably calls a quiet world quiet: final control detection 7/10 runs vs 10/10 for all Anthropic arms) at the cost of under-detecting real faults (60% detection – the only arm to miss genuine anomalies outright).

Finding 4: Anomaly dating fails in two decomposable ways – and unconditional date commitment is the invariant

Judged anomaly-onset dates in gravity worlds (truth: day 12; Figure 3):

```
 B1  haiku:      4, 6, 7, 7, 7, 8, 9, 9, 10,10,10,10,10,10, 11, 12, 18, 21, 23
B2  sonnet:     8, 8, 9, 10,10,10,10,10,10, 11×8, 16, 22, 28
B3a sonar-pro:  6, 8, 8, 8, 10,10,10,10, 11, 12, 24
B3b sonnet-nmp: 8, 8, 9, 10,10,10,10, 11×8, 12, 15, 24
```

Summary statistics (dated agents only): median error −2 days (B1), −1.5 (B2), −2 (B3a), −1 (B3b); estimates preceding day 12: 15/19, 17/20, 9/11, 15/18; within ±1 day of truth: 2/19, 8/20, 2/11, 9/18. Every arm also committed to "onset" dates in control worlds where nothing happened (16, 20, 11, and 20 of 20 agents respectively).

Control analysis (evidence vs interpretation). To test whether the early bias resides in the agents or in the evidence itself, we ran a maximum-deviation change-point estimator on the *exact per-agent observation streams* from the B1 and B2 event logs. On haiku's streams the detector estimates median day 10.5 with 13/20 before day 12 – closely matching haiku's own dating – and reaches $z \geq 3$ in only 8/20 cases: haiku's erratic, self-chosen measurement schedules genuinely blur the onset, so its dating error is substantially evidence-side and self-inflicted. On sonnet's more systematic streams the detector estimates median day 12.0 (11/20 within 11–13), while sonnet itself reported a mode of 10–11: a residual ≈1-day interpretation-side early bias beyond what its own evidence explains. In control worlds, the detector's maximum-deviation "onsets" never reached significance (0/39 at $z \geq 3$) – yet agents committed to onset dates anyway. The behaviour that survives decomposition across all tested model and prompt conditions is therefore not mis-estimation per se but unconditional commitment: agents date the strongest apparent change point without applying any significance criterion, in quiet and perturbed worlds alike. Whether the shared agent architecture (observation presentation, notebook statistics, memory construction) contributes causally remains untested and is registered as future work.

6. Discussion

The ceiling is not one failure. The haiku/sonnet contrast (generation failure vs commitment failure, Finding 1) indicates that different models reach the same endpoint through different cognitive trajectories – the ceiling is a shared endpoint reached by different mechanisms. Any account of its origin must explain both routes, and interventions that fix one (hypothesis-generation scaffolding) may do nothing for the other (commitment support).

Where does the physics ceiling come from? Study 1 weakens two simple explanations: insufficient model capability (the ceiling survives a tier jump, and capability demonstrably improves every adjacent skill) and the explicit prompt prior (it survives the ablation). What remains: shared training-era epistemic conservatism across contemporary models; the agent architecture and its evidence presentation; the experimental framing (30 days, two agents, one discriminating instrument); or interactions among these. We do not claim to have localised it – only to have made the two easiest stories untenable.

The calibration trade-off as a design axis. Deployed agent systems will inherit some version of the mundanity prior, explicitly or through training. Our results suggest it functions as a genuine calibration dial with costs in both positions, and that "remove the guardrail" does not convert conservatism into insight – it converts it into false alarms.

Model choice as epistemic-culture choice. For multi-agent systems, the selection of a foundation model appears to set collaboration rate, blindness discipline, fabrication propensity, and anomaly appetite more strongly than it sets diagnostic accuracy. Heterogeneous societies – where a fabricating, hyper-social model coexists with a silent, well-grounded one – are the natural next experiment (Study 2, with an in-world publication mechanism, will test whether social scale corrects or amplifies individual failures, and whose claims propagate).

7. Limitations

1. **n = 10 per condition per arm.** Rates carry wide intervals (binary outcomes ±~30pp); the headline 0/40 is robust in direction but its upper bound is ~7.5%, not zero. Study 1 is exploratory by design.
2. **Judge provenance.** All judged metrics use an Anthropic judge over Anthropic and Perplexity agents. For cross-lab claims we triangulated with the model-free lexicon tripwire and a manual claim audit; full inter-judge agreement (a second, non-Anthropic judge over a sample) is registered as future work.
3. **Communication-volume confound** in the confabulation dissociation (§5, Finding 3), to be resolved by budget-matched runs.
4. **Pretraining contamination, by design.** Agents know Bostrom, Kuhn, and The Matrix from pretraining. Observer Zero does not test whether naive minds can invent the simulation hypothesis; it tests how LLM personas reason under controlled epistemic conditions, with fictional constants ensuring the *world's state* cannot be retrieved from memory.
5. **Metric breadth.** eval-v2's detection criterion counts self-error and environmental attributions as anomaly mass, inflating "detection" for chatty models relative to eval-v1; both definitions are reported in the artifact and all arms are scored identically.
6. **Single architecture.** All results are within one agent loop, memory design, and prompt family. Finding 4 in particular is architecture-invariant only in the weak sense of "invariant across what we varied."
7. **No model-side sampling seed** exists on any provider; run-level variance includes sampling variance (this is also the phenomenon).

8. Reproducibility

All 150 run artifacts contain: complete event logs with ground truth, every model call (prompt, completion, tokens, cost, latency, prompt version), belief timelines, memories, replication episodes, leak-audit results, evaluator outputs with judge calls, and the frozen manifest. World randomness is fully seeded and order-independent; the mock arm reproduces bit-identically. Repository: engine, runner, battery, and evaluator are ~70 tests, TypeScript, with the full battery pipeline runnable for $0 via the mock provider. Repository: https://github.com/nickjlamb/observer-zero.

References

[1] Park JS, O'Brien JC, Cai CJ, Morris MR, Liang P, Bernstein MS. Generative agents: Interactive simulacra of human behavior. In: Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23). New York, NY: Association for Computing Machinery; 2023. doi:10.1145/3586183.3606763. arXiv:2304.03442. [2] Vezhnevets AS, Agapiou JP, Aharon A, et al. Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. arXiv [Preprint]. 2023. arXiv:2312.03664. [3] Lu C, Lu C, Lange RT, Foerster J, Clune J, Ha D. The AI Scientist: Towards fully automated open-ended scientific discovery. arXiv [Preprint]. 2024. arXiv:2408.06292. [4] Ji Z, Lee N, Frieske R, et al. Survey of hallucination in natural language generation. ACM Computing Surveys. 2023;55(12):1–38. doi:10.1145/3571730. [5] Kadavath S, Conerly T, Askell A, et al. Language models (mostly)

know what they know. arXiv [Preprint]. 2022. arXiv:2207.05221. [6] Kuhn TS. The Structure of Scientific Revolutions. Chicago, IL: University of Chicago Press; 1962. [7] Tversky A, Kahneman D. Judgment under uncertainty: Heuristics and biases. Science. 1974;185(4157):1124–1131. doi:10.1126/science.185.4157.1124. [8] Bostrom N. Are we living in a computer simulation? The Philosophical Quarterly. 2003;53(211):243–255. doi:10.1111/1467-9213.00309.

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Generative AI systems were used during study design, software implementation, analysis, and manuscript preparation. One system was used primarily for critical review of experimental design and interpretation, and another for implementation and analysis assistance. All experimental decisions, validation, interpretation, and final manuscript content were reviewed and approved by the author. The AI systems are not authors.